

Cloud Computing Project 2 - GCP

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1) What challenges do you think will be faced if you deploy it on the premise data center?

A) The challenges to be faced while deploying it on the premise data center are:

1. Operational cost investment.
2. Maintenance of the host.
3. Scalability of the resources.
4. Disaster management.

2) List the advantages or improvements for implementation of cloud?

A) Advantages of deploying an application on cloud are:

1. Various service models and offerings are available.
2. Geographical coverage.
3. Deployment simplicity
4. Customization.
4. Reliability.

3) Using Compute engine service on GCP, deploy your application. Note the process, challenges you faced. You need to know that Compute Engine is an IaaS offering on GCP?

A)

1. Deploy Compute Engine
2. Configure Ports and networking part
3. login with SSH, install and start web server
4. place code files in var/www/html
5. check for all the services and ports.

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Create an instance

To create a VM instance, select one of the options:

- New VM instance**
Create a single VM instance from scratch
- New VM instance from template**
Create a single VM instance from an existing template
- New VM instance from machine image**
Create a single VM instance from an existing machine image
- Marketplace**
Deploy a ready-to-go solution onto a VM instance

Name *
gcinstance-3

Labels
+ ADD LABELS

Region *
us-central1 (Iowa)

Zone *
us-central1-a

Region is permanent Zone is permanent

Machine configuration

Machine family
GENERAL-PURPOSE COMPUTE-OPTIMIZED MEMORY-OPTIMIZED GPU

Machine types for common workloads, optimized for cost and flexibility

Series
E2

Monthly estimate
\$25.46
That's about \$0.03 hourly
You have \$287.58 free trial credits remaining
Pay for what you use: No upfront costs and per second billing

▼ DETAILS

CLOUD SHELL
Terminal (silent-fuze-341016) x +

```
Unexpected attribute "runtime" for object of type IndexDefinitions.  
in "/home/gajjelasuryateja2015/cloudshell_open/App01/index.yaml", line 2, column 10  
gajjelasuryateja2015@cloudshell:~/cloudshell_open/App01 (silent-fuze-341016)$ gcloud app deploy index.yaml  
ERROR: (gcloud.app.deploy) An error occurred while parsing file: [/home/gajjelasuryateja2015/cloudshell_open/App01/index.yaml]  
Unexpected attribute "runtime" for object of type IndexDefinitions.  
in "/home/gajjelasuryateja2015/cloudshell_open/App01/index.yaml", line 2, column 10  
gajjelasuryateja2015@cloudshell:~/cloudshell_open/App01 (silent-fuze-341016)$
```

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Compute Engine

VM instances

Virtual machines

- VM instances
- Instance templates
- Sole-tenant nodes
- Machine images
- TPUs
- Committed use discounts
- Marketplace
- Release Notes

INSTANCES INSTANCE SCHEDULE

VM instances are highly configurable virtual machines for running workloads on Google infrastructure. [Learn more](#)

Filter Enter property name or value

Status	Name	Zone	Machine type	Recommendations	Connect
☒	ucinstance-01	us-central1-a	e2-medium		SSH

Select an instance

PERMISSIONS LABELS MONITORING

Please select at least one resource.

CLOUD SHELL
Terminal (silent-fuze-341016) x +

Open Editor

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Compute Engine

ucinstance-01 EDIT RESET CREATE MACHINE IMAGE

Virtual machines

- VM instances
- Instance templates
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DETAILS OBSERVABILITY OS INFO SCREENSHOT

SSH CONNECT TO SERIAL CONSOLE

Connecting to serial ports is disabled

Logs
Cloud Logging
Serial port 1 (console)
SHOW MORE

Basic information

Name	ucinstance-01
Instance Id	1347411435254669689
Description	None
Type	Instance

CLOUD SHELL
Terminal (silent-fuze-341016) x +

```
gajjelasuryateja2015@ucinstance-01: /var/www/html$  
gajjelasuryateja2015@ucinstance-01:~$ cd ..  
gajjelasuryateja2015@ucinstance-01:~/home$ ls  
gajjelasuryateja2015@ucinstance-01:~/home$ cd gajjelasuryateja2015/  
gajjelasuryateja2015@ucinstance-01:~/home$ ls  
gajjelasuryateja2015@ucinstance-01:~$ cd ..  
gajjelasuryateja2015@ucinstance-01:~/home$ cd ..  
gajjelasuryateja2015@ucinstance-01:/$ ls  
bin dev home lib32 libx32 media opt root sbin sys usr  
boot etc lib lib64 lost+found mnt proc run srv var  
gajjelasuryateja2015@ucinstance-01:/$ cd var/www/html  
gajjelasuryateja2015@ucinstance-01:var/www/html$ ls  
index.html test01.txt  
gajjelasuryateja2015@ucinstance-01:var/www/html$
```

CLOUD SHELL
Terminal (silent-fuze-341016) x +

```
Unexpected attribute "runtime" for object of type IndexDefinitions.  
in "/home/gajjelasuryateja2015/cloudshell_open/App01/index.yaml", line 2, column 10  
gajjelasuryateja2015@cloudshell:~/cloudshell_open/App01 (silent-fuze-341016)$ gcloud app deploy index.yaml  
ERROR: (gcloud.app.deploy) An error occurred while parsing file: [/home/gajjelasuryateja2015/cloudshell_open/App01/index.yaml]  
Unexpected attribute "runtime" for object of type IndexDefinitions.
```

```
gajelasuryateja2015@ucinstance-01: ~
debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Wed Feb 16 16:23:29 2022 from 35.235.244.32
gajelasuryateja2015@ucinstance-01:~$ ls
gajelasuryateja2015@ucinstance-01:~$ cd ..
gajelasuryateja2015@ucinstance-01:/home$ ls
gajelasuryateja2015  suryatejagajjela
gajelasuryateja2015@ucinstance-01:/home$ cd ..
gajelasuryateja2015@ucinstance-01:/home$ ls
bin  dev  home  lib32  libx32  media  opt  root  sbin  sys  usr
boot  etc  lib  lib64  lost+found  mnt  proc  run  srv  tmp  var
gajelasuryateja2015@ucinstance-01:/home$ cd var/www/
gajelasuryateja2015@ucinstance-01:/var/www$ ls
html
gajelasuryateja2015@ucinstance-01:/var/www$ cd html
gajelasuryateja2015@ucinstance-01:/var/www/html$ vim html
gajelasuryateja2015@ucinstance-01:/var/www/html$ ls
index.html  test01.txt
gajelasuryateja2015@ucinstance-01:/var/www/html$ sudo vim index.html
gajelasuryateja2015@ucinstance-01:/var/www/html$ sudo vim index.html
gajelasuryateja2015@ucinstance-01:/var/www/html$ sudo nano index.html
gajelasuryateja2015@ucinstance-01:/var/www/html$ sudo nano index.html
gajelasuryateja2015@ucinstance-01:/var/www/html$ Connected, host's fingerprint: ssh-rsa 0 1A:B1:B9:26:5A:0D:5B:DA:37:93
3:4E:AB:0B:80:CE:8F:8C:15:35:64:96:6F:87:18:BA:2B:FG:8F:17:FB:C6
linux ucinstance-01 4.19.0-18-cloud-amd64 #1 SMP Debian 4.19.208-1 (2021-09-29) x86_64

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
Last login: Wed Feb 16 17:37:22 2022 from 35.235.244.34
gajelasuryateja2015@ucinstance-01:~$
```

URL: <http://35.224.241.11/>

Google Chrome - Feb 16, 12:34

STUDENT CREDIT APPROVAL CHECK

FIRST NAME (max. 30 characters a-z and A-Z)

LAST NAME (max. 30 characters a-z and A-Z)

DATE OF BIRTH / /

EMAIL ID

MOBILE NUMBER (10 digit number)

ADDRESS

CITY (max. 30 characters a-z and A-Z)

PIN CODE (6 digit number)

COUNTRY

CREDIT DATA

SL.No.	Year	Annual Income	Loan debt
1	Previous Year	40000	10000
2	Current Year	35000	5000

(10 char max) (up to 2 decimal)

Enter Your Credit Score

Enter Credit Amount Needed(Max 10k USD)

STUDENT CREDIT APPROVAL CHECK

35.224.241.11 says
Your Current Capacity is NOT Eligible to get Credit

OK

FIRST NAME
LAST NAME
DATE OF BIRTH
EMAIL ID
MOBILE NUMBER
ADDRESS
CITY
PIN CODE
COUNTRY
CREDIT DATA

SL.No.	Year	Annual Income	Loan debt
1	Previous Year	40000	10000
2	Current Year	20000	5000

Enter Your Credit Score
Enter Credit Amount Needed(Max 100k USD)

Evaluate

4) Do you choose App Engine Stand Environment or App Engine Flexible as deployment options for your application? Why?

A) We have chosen App Engine Flexible as deployment options for your application. The App Engine flexible environment scales your app up and down automatically while balancing load. Also, another reason to choose is that the application instances run within docker containers on Compute Engine virtual machines (VM). flexible environment is optimal for applications with Source code that is written in a version of any of the supported programming languages like Python, Java, Node.js, Go, Ruby, PHP, or .NET easy when uses or depends on frameworks that include native code and Accesses the resources or services of your Google Cloud project that reside in the Compute Engine network which can avoid sudden and extreme spikes of traffic which require immediate scaling.

5) Deploy your application on App Engine and then compare and comment on your experience with Compute Engine based deployment?

Point: When we compared application deployment in compute Engine to App Engine, Deploying in App Engine is easier and flexible. We can skip deploying a virtual machine, VM configurations and networking settings if we are deploying an application in App Engine. In case of continuous Integration and continuous Deployment, App Engine is super useful for developers

we can just commit and pull the new code from repos so that we can directly update the new version directly without the downtime for application.

<http://34.133.53.39/>

The screenshot shows the Google Cloud Platform App Engine dashboard. The left sidebar contains navigation links: Dashboard, Services, Versions, Instances, Task queues, Cron jobs, and Release Notes. The main content area displays a 'Welcome to App Engine' message with a green checkmark indicating that the application has been created. Below the message is a 'GET STARTED' button. At the bottom, a terminal window is open, showing the following commands and output:

```
gajjelasuryateja2015@cloudshell:~ (silent-fuze-341016) $ gcloud app browse
Did not detect your browser. Go to this link to view your app:
https://silent-fuze-341016.uc.r.appspot.com
gajjelasuryateja2015@cloudshell:~ (silent-fuze-341016) $ gcloud beta app instances list
```

The screenshot shows the 'Add a new custom domain' page in the Google Cloud Platform App Engine dashboard. The page provides instructions on how to point a domain to the App Engine app. It includes a list of domains to choose from, a section for mapping domains and subdomains, and a 'SAVE MAPPINGS' button. The terminal window at the bottom shows the following commands and output:

```
gajjelasuryateja2015@cloudshell:~ (silent-fuze-341016) $ gcloud app browse
Did not detect your browser. Go to this link to view your app:
https://silent-fuze-341016.uc.r.appspot.com
gajjelasuryateja2015@cloudshell:~ (silent-fuze-341016) $ gcloud beta app instances list
```

URL: <https://silent-fuze-341016.uc.r.appspot.com/>

STUDENT CREDIT CHECK

FIRST NAME (max 30 characters a-z and A-Z)

LAST NAME (max 30 characters a-z and A-Z)

DATE OF BIRTH (max 30 characters a-z and A-Z)

EMAIL ID

MOBILE NUMBER (10 digit number)

ADDRESS

CITY (max 30 characters a-z and A-Z)

PIN CODE (5 digit number)

STATE (max 30 characters a-z and A-Z)

COUNTRY

INCOME DETAILS

SL.No.	Income	Total Debt Amount
1	previous year 40000	70000
2	current year 50000	45000

STUDENT CREDIT CHECK

34.133.53.39 says
Your current Capacity is NOT Eligible to get Credit

FIRST NAME (max 30 characters a-z and A-Z)

LAST NAME (max 30 characters a-z and A-Z)

DATE OF BIRTH (max 30 characters a-z and A-Z)

EMAIL ID

MOBILE NUMBER (10 digit number)

ADDRESS

CITY (max 30 characters a-z and A-Z)

PIN CODE (5 digit number)

STATE (max 30 characters a-z and A-Z)

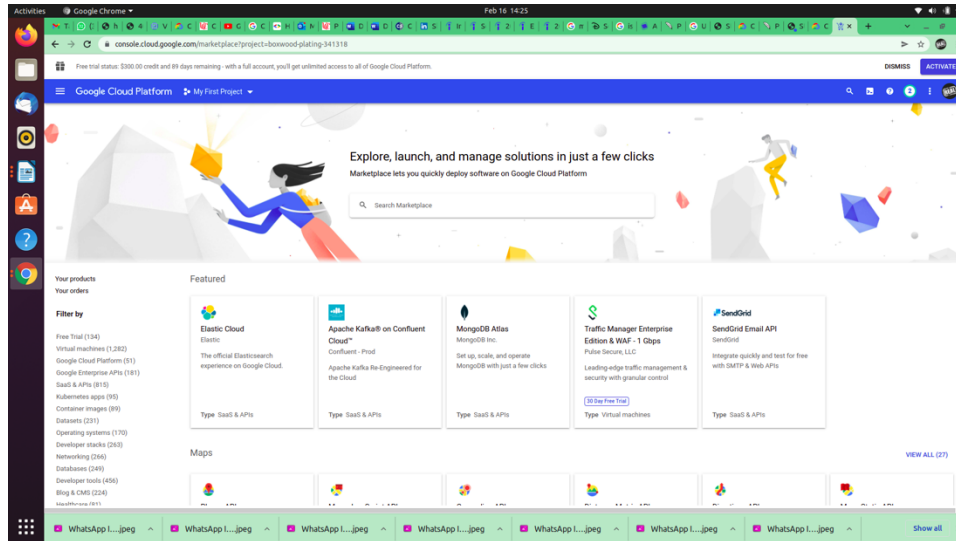
COUNTRY

INCOME DETAILS

SL.No.	Income	Total Debt Amount
1	previous year 40000	70000
2	current year 50000	45000

6) Explore "Marketplace" from GCP and identify any ready to use components supporting your project?

A)



The components supporting the deployment are:

There are few very useful components like cloud Run, app engine.

1) Cloud Run with Firebase Hosting to generate and serve your dynamic content or build REST APIs as microservice which is very useful and easy to deploy most of the supportive platforms.

2) Data Base components such as Mongo DB, PostGre SQL and 'N' number of APIs

7) Comment and conclude your way of deployment?

A) We deployed our application using Compute Engine, and App Engine and in this document, we have given a brief description of the process, the challenges we faced during the deployment. As proof, we have posted URLs and screenshots showing our results. We have also explored the GCP Market place and identified ready to use database components such as Mongo DB, Post-GRE-SQL.

We tried deploying using Cloud Run and it's very easy to deploy our application from GitHub, also we can add Domain and our subdomain for web URL, Container Deployment like Kubernetes and Docker are also very useful to deploy the applications and provide scalable resources, which are light weight to run the applications and Databases.